

Assignment 3: Data Analytics

188.429 Business Intelligence (VU 4,0) – WS 2025

Group 13

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1 Business Understanding (Student A + B contributed)

1.1 Data Source and Scenario

The dataset used in this project is the Ames Housing Dataset, originating from residential home sales in Ames, Iowa, USA. It contains 2,930 instances and 79 features describing physical property characteristics such as lot dimensions, building quality, neighborhood, construction year, and interior attributes. The target variable is **SalePrice**, representing the final selling price of each home.

For this assignment, we assume the role of a real-estate analytics company aiming to support property valuation workflows. Each row corresponds to a single house sale, and the diverse feature set makes it suitable for regression modeling and real-world price prediction tasks.

1.2 Business Objectives

The business objective is to predict residential property sale prices accurately to support real-estate valuation and decision-making. The model will help reduce manual appraisal time, provide consistent price estimates, and support identification of overvalued or undervalued properties.

Non-goals: We do not forecast future market trends, buyer behavior, or housing demand. We also do not perform demographic analysis unrelated to property characteristics.

1.3 Business Success Criteria

Business success criteria include: 1) Reduction of manual appraisal time by 20% through automated pre-valuation. 2) Reduction of valuation deviation from expert appraisers by at least 10%. 3) Increased consistency of valuations across agents, reducing internal price variance by 15%.

Stakeholders include real-estate valuation analysts, acquisition teams, and management.

1.4 Data Mining Goals

The goal is to build a regression model that predicts **SalePrice** using the most relevant features of the Ames dataset. This includes evaluating all variables, identifying the most informative ones, and removing irrelevant or redundant features to improve model accuracy and interpretability. The model will be trained, tuned, and compared across several algorithms while ensuring reproducibility and avoiding data leakage.

1.5 Data Mining Success Criteria

Success is defined as achieving: - $RMSE \leq 35,000$ or $MAE \leq 20,000$ on the test set, - $R^2 \geq 0.80$, - A reduced and meaningful feature set that retains or improves predictive performance, - Stable results across different random splits, - Full reproducibility of preprocessing, feature selection, and model training steps.

1.6 AI Risk Aspects

AI risk aspects include location-based bias because neighborhood variables may reflect socioeconomic patterns. The model could unintentionally reproduce valuation disparities between neighborhoods.

This model is categorized as “Limited Risk” under EU AI guidelines, since it supports but does not automate legally binding decisions. Ethical mitigation includes monitoring errors across neighborhoods, ensuring human oversight in final valuations, and avoiding overreliance on sensitive location features.

2 Data Understanding (Student A contributed)

2.1 (2a) Attribute types, units of measurement, and semantics

The Ames Housing dataset contains mixed attribute types describing residential properties. Numerical variables capture measurable quantities such as areas in square feet (e.g., **GrLivArea**, **TotalBsmtSF**, **1stFlrSF**), counts

(e.g., `GarageCars`, `FullBath`), and years (e.g., `YearBuilt`, `GarageYrBlt`, `YearRemodAdd`). The target variable `SalePrice` is a continuous value in USD. Categorical variables encode property categories and market segmentation (e.g., `Neighborhood`, `GarageType`, `Fence`). Several quality-related fields are ordinal ratings (e.g., `OverallQual`, `KitchenQual`), where higher categories indicate better condition/quality. Semantically, the dataset reflects size, quality, location, and amenity information that jointly determine sale prices.

2.2 (2b) Statistical properties describing the dataset, including correlations

The distribution of `SalePrice` is right-skewed, with a majority of houses concentrated in a mid-price range and a smaller set of high-priced properties forming a long tail. Correlation analysis on numeric variables indicates that `SalePrice` is positively associated with both size and quality features. In particular, `OverallQual` and `GrLivArea` show strong positive associations with `SalePrice`. Several predictors are also correlated with each other (e.g., multiple square-footage variables), implying multicollinearity, which motivates the use of regularization in linear models. Figure 1 summarizes correlations among key numeric features.

2.3 (2c) Data quality aspects: missingness, plausibility, outliers, provenance

Table 1: Examples of missingness and data quality issues (Ames Housing)

Attr.	Miss.%	Type	Implication
<code>PoolQC</code>	99.6	Cat.	Mostly “no pool” (not random).
<code>MiscFeature</code>	96.4	Cat.	Rare feature; missing often encodes absence.
<code>Alley</code>	93.2	Cat.	Many properties have no alley access.
<code>Fence</code>	80.5	Cat.	Missing is meaningful (no fence).
<code>MasVnrType</code>	60.6	Cat.	Linked to construction; may correlate with price.
<code>FireplaceQu</code>	48.5	Ord.	Often indicates no fireplace.
<code>LotFrontage</code>	16.7	Num.	Measurement gaps; imputation needed.
<code>GarageYrBlt</code>	5.4	Num.	Missing for houses without garages.

Table 1 illustrates that missingness is highly uneven and frequently semantically meaningful. Extremely high missingness in variables such as `PoolQC`, `MiscFeature`, `Alley`, and `Fence` is plausibly explained by non-applicability (most houses do not have these amenities), rather than random data loss. Moderate missingness in variables such as `LotFrontage` may stem from incomplete records or measurement gaps. Missingness can affect model performance and can introduce bias if it is correlated with location or price. The dataset also contains outliers, including very high `SalePrice` values (Figure 4) and unusually large `GrLivArea` observations (Figure 3). These points can disproportionately influence regression training and evaluation metrics such as RMSE. Therefore, decisions about missingness handling, feature removal, and outlier handling must be made in a leakage-free way (fit/decide on TRAIN only after splitting) and documented for reproducibility.

2.4 (2d) Visual exploration of data properties and hypotheses

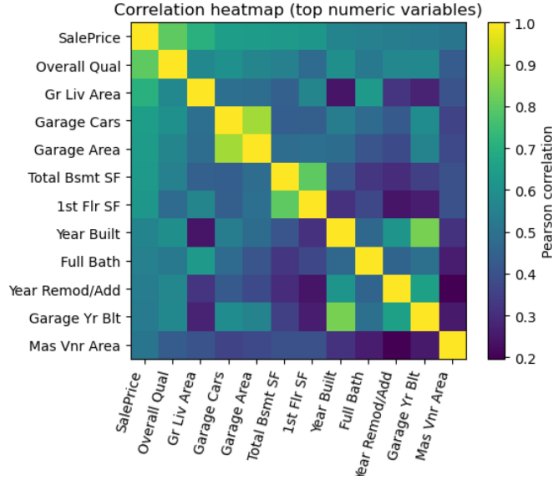


Figure 1: Correlation heatmap (top numeric variables).

Figure 1 indicates strong positive associations between `SalePrice` and `OverallQual` / `GrLivArea`, and multicollinearity among size-related predictors.

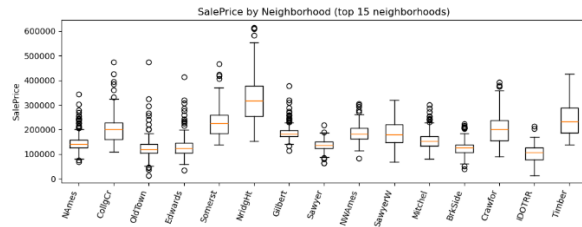


Figure 2: `SalePrice` by `Neighborhood` (top neighborhoods).

Figure 2 shows clear price segmentation by neighborhood, motivating subgroup-based evaluation.

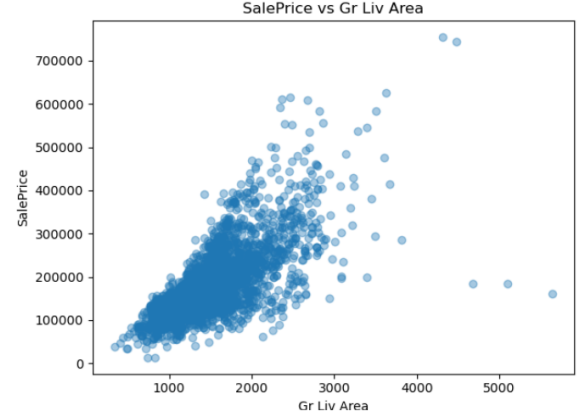


Figure 3: `SalePrice` vs `GrLivArea`.

Figure 3 supports a strong positive relationship between living area and price, with a small number of atypical extreme cases.

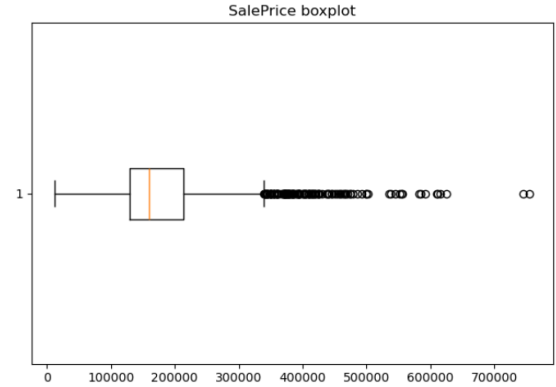


Figure 4: `SalePrice` boxplot.

Figure 4 confirms right-skewness and high-end outliers, explaining why RMSE can be sensitive to rare expensive properties.

Based on these plots, the following hypotheses were formed: `OverallQual` and `GrLivArea` are expected to be among the most predictive features; neighborhood encodes strong market segmentation; `SalePrice` outliers likely affect error-based metrics; correlated size variables imply redundancy that can be managed via regularization.

2.5 (2e) Ethically sensitive attributes, minority groups, imbalance, and bias-relevant distributions

The dataset does not contain explicit protected attributes (e.g., race, gender). However, **Neighborhood** and other location-related variables can act as socioeconomic proxies and may reflect structural inequalities present in housing markets. Some neighborhoods may have few observations, which creates minority or under-represented subgroups and can lead to unstable subgroup error estimates. For evaluation, this motivates reporting overall metrics together with subgroup metrics and interpreting disparities in the context of sample sizes.

2.6 (2f) Potential risks and additional types of bias; expert questions

Potential risks include proxy bias through **Neighborhood**, missing-not-at-random patterns where missingness correlates with price or area, and sampling bias if the dataset underrepresents certain market segments. Additional concerns include temporal bias if sales span different market conditions. Questions for an external expert or data owner include: what missingness represents for high-missing features (true unknown vs non-applicable); whether any neighborhoods are systematically underreported; whether quality ratings are consistently assessed; and whether the dataset represents a single stable market period or includes structural shifts.

2.7 (2g) Likely required actions for data preparation

The analysis suggests the need for leakage-free splitting before preprocessing; explicit handling of high-missing features (removal or semantic encoding of “None”); imputation of remaining missing values using TRAIN-only statistics; encoding of categorical variables and scaling of numeric variables fit on TRAIN only; and a documented outlier strategy applied on TRAIN only.

In addition, subgroup evaluation by neighborhood should be included to monitor potential disparities in model errors.

3 Data Preparation (Student B contributed)

3.1 (a) Data Preprocessing

All preprocessing steps were designed to be reproducible and leakage-free. We perform an IID random split with `seed=42` into TRAIN/VAL/TEST before any data preparation. All decisions (dropped columns, imputers, encoders, scaler) are fitted on TRAIN only and then applied unchanged to VAL/TEST.

Columns were excluded based on missingness on TRAIN using the rule `missing_pct > 40%`. Under this rule, the following variables were removed from the feature set due to extreme or highly uneven missingness: **PoolQC**, **MiscFeature**, **Alley**, **Fence**, **MasVnrType**, **FireplaceQu**. Remaining missing numeric values were imputed with the TRAIN median and categorical values with the TRAIN mode. Categorical variables were one-hot encoded with `handle_unknown=ignore`. Numeric features were standardized using **StandardScaler**. Both transformations were fitted on TRAIN and applied to VAL/TEST.

3.1.1 Train/Validation/Test Split

We split the dataset into training/validation/test sets BEFORE any data preparation to prevent data leakage. All cleaning/feature engineering steps were fitted on the training set only and then applied to validation/test. Random IID split with `seed=42`, `test_size=0.2`, `val_size=0.2` (fractions of full dataset).

3.1.2 Outlier Handling (TRAIN only)

Outlier removal is applied ONLY to the training set after data splitting to prevent data leakage. Rows were identified during the Data Understanding phase and removed according to

the documented outlier decision. Rule: rows flagged as outliers in ≥ 2 numeric columns.

3.1.3 Missingness After Outlier Removal (Train)

Table 2: Top missing columns after outlier handling (train only)

Column	Missing	Missing %
Pool QC	1727	99.37%
Misc Feature	1669	96.03%
Alley	1634	94.02%
Fence	1378	79.29%
Mas Vnr Type	1027	59.09%
Fireplace Qu	848	48.79%
Lot Frontage	279	16.05%
Garage Qual	96	5.52%
Garage Cond	96	5.52%
Garage Type	96	5.52%

Table 2 confirms that high-missingness attributes remain concentrated in a small set of mostly non-applicable features, motivating the explicit missingness-based removal rule and targeted imputation.

3.1.4 Missing Value Handling Decision

Decision after inspecting missingness report (train only): exclude columns with missing_pct $> 40.0\%$. Impute numeric with median and categorical with mode. Fit statistics on TRAIN only; apply to VAL/TEST to avoid leakage.

3.1.5 Feature Engineering Decision

Numeric features scaled using `StandardScaler` fitted on the training set only. Categorical features encoded using `OneHotEncoder` with `handle_unknown=ignore`, fitted on the training set only and applied to validation and test sets. Target variable excluded from feature engineering.

3.2 (b) Considered but not applied

Additional preprocessing steps were considered but not applied in this interim version. A log-transform of `SalePrice` could reduce skewness but would complicate interpretation and back-transformation of errors in USD. More complex outlier strategies such as winsorization or robust regression were not used to keep the pipeline simple and fully reproducible. Binning continuous variables was not applied because it can discard information and is not necessary for Ridge. More aggressive feature selection methods were deferred to maintain transparency and because regularization already mitigates multicollinearity.

3.3 (c) Options and potential for derived attributes

Potential derived attributes include property age computed from `YearBuilt`, a remodel indicator derived from `YearRemodAdd` relative to `YearBuilt`, total area aggregates combining basement and above-ground measures, and interaction terms between quality and size (e.g., `OverallQual` multiplied by `GrLivArea`). These features may improve predictive performance and interpretability, but they are documented as options rather than applied in this interim report to keep the baseline pipeline minimal.

3.4 (d) Options for additional external data sources

External data that could strengthen valuation accuracy and business relevance includes time-indexed market indicators (interest rates, housing indices), enriched location attributes (distance to amenities, schools, transit, crime rates), comparable-sales information, and renovation or energy-efficiency records. These sources are not integrated here, but they are plausible extensions if the objective shifts from baseline modeling toward production valuation support.

4 Modeling (Student A contributed)

4.1 (a) Algorithm selection

Context from earlier phases: we perform an IID train/validation/test split with seed=42 BEFORE any preparation to prevent leakage, then apply `OneHotEncoder(handle_unknown=ignore)` for categorical features and `StandardScaler` for numeric features fitted on TRAIN only and applied to VAL/TEST. This yields a high-dimensional, sparse feature space typical for tabular housing data. The modeling goal is accurate `SalePrice` prediction with reproducibility and stable generalization; the system is used as decision support (Limited Risk), so transparent and explainable behavior is preferred.

Suitable regression algorithms considered: (1) Regularized linear models (Ridge/Lasso/ElasticNet): strong baselines for sparse one-hot encoded data; fast to train, reproducible, and interpretable via coefficients; regularization controls overfitting in high-dimensional spaces. (2) Tree-based ensembles (RandomForest / Gradient Boosting): can capture non-linearities and interactions, but require substantially more tuning effort, are less transparent, and can increase compute cost; also less aligned with the explicit scaling + OHE pipeline already defined. (3) Simple baseline (LinearRegression): useful as reference, but may overfit in a high-dimensional sparse setting without regularization.

Final selection: Ridge regression is selected as the primary model for experiments because it matches the defined preprocessing (scaled numeric + sparse OHE), provides stable performance in high-dimensional settings via L2 regularization, is computationally efficient for repeated validation runs, and improves interpretability and auditability of a valuation-support model. Non-linear ensembles remain candidates for later extension if Ridge performance is insufficient to reach the target thresholds, but Ridge is the most suitable choice for

the required reproducible tuning experiment in this assignment.

4.2 (b) Hyper-parameter selection and tuning plan

Ridge regression exposes the key hyperparameter `alpha` (L2 regularization strength), plus secondary options such as solver, `max_iter`, and tolerance. The most relevant tuning parameter is `alpha` because it directly controls the bias-variance trade-off and prevents overfitting in the high-dimensional sparse feature space produced by one-hot encoding. We tune `alpha` on a small explicit grid to balance compute effort and reproducibility. We select a log-spaced set of candidate values to cover weak-to-strong regularization. All tested values are documented explicitly.

4.3 (c) Train/validation/test split definition (reproducible)

We use a fixed IID random split with seed=42. The dataset is split into TRAIN/VAL/TEST before any preprocessing to prevent leakage. All preprocessing steps are fitted on TRAIN only. Hyperparameter selection (Ridge `alpha`) is performed using VAL only. The TEST set remains untouched until the final evaluation step and is used exactly once for the final reported performance. This ensures a clean separation between model selection and final generalization assessment.

4.4 (d) Training and validation (explicit settings tested)

All Ridge models were trained using the same fixed preprocessing pipeline: numeric features were standardized using `StandardScaler` fitted on TRAIN only, categorical features were encoded using `OneHotEncoder(handle_unknown=ignore)` fitted on TRAIN only, and the resulting design matrix was used for Ridge regression. The Ridge model used `fit_intercept=True` and the default solver (`solver=auto`). The only tuned hyperparameter was the regularization

strength α . We tested the explicit grid $\{0.01, 0.1, 1, 10, 100, 1000\}$, trained on TRAIN, and evaluated on VAL using RMSE, MAE, and R^2 computed in the original USD scale of SalePrice.

Table 3: Ridge tuning results on VAL (explicit alpha grid).

α	RMSE	MAE	R^2
0.01	23795.288	16409.699	0.90351
0.1	23851.936	16438.318	0.90305
1	24320.440	16779.586	0.89920
10	24354.283	16532.429	0.89892
100	24152.661	16388.573	0.90058
1000	26994.950	18407.853	0.87581

Table 3 shows that weak regularization (alpha around 0.01–0.1) achieves the lowest RMSE and highest R^2 . When alpha becomes very large (e.g., 1000), RMSE increases and R^2 decreases, indicating underfitting due to excessive coefficient shrinkage in a high-dimensional feature space. We therefore select the alpha with the lowest validation RMSE and retrain the final model on TRAIN+VAL before performing a single holdout TEST evaluation.

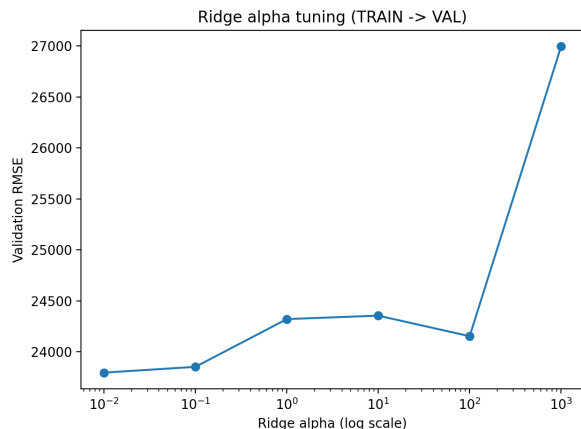


Figure 5: Validation RMSE across Ridge alpha values (log scale).

4.5 (f) Model selection decision

Best model selected on validation: Ridge(alpha=0.01) using lowest validation RMSE. Final model retrained on TRAIN+VAL

with identical hyperparameters and evaluated once on the untouched TEST set; final holdout test results are recorded as RMSE=22731.376, MAE=16087.706, R^2 =0.90773.

4.6 (g) Final model training and holdout test evaluation

Holdout TEST evaluation for final Ridge(alpha=0.01) trained on TRAIN+VAL: RMSE=22731.376, MAE=16087.706, R^2 =0.90773.

5 Evaluation (Student B contributed)

5.1 (a) Final model performance on TEST

Final model: Ridge(alpha=0.01) trained on TRAIN+VAL.

Table 4: Final model performance on TEST

Metric	Value	Notes
RMSE	22,689.09	lower is better
MAE	16,062.85	lower is better
R^2	0.9081	higher is better

Table 5: MAE by SalePrice quantile (TEST)

Quantile	MAE
1	13,868.87
2	12,556.59
3	15,581.43
4	14,621.36
5	23,699.97

Table 4 shows strong overall test performance. Table 5 indicates the highest MAE in the top price quantile, consistent with higher variance and fewer high-priced examples. Protocol: hold-out TEST evaluation (single split). Hyperparameter alpha selected on VAL only; TEST untouched during tuning.

5.2 (b) Benchmarks and baselines

5.2.1 (i) State-of-the-art performance references

State-of-the-art performance for the Ames Housing dataset is commonly reported in public ML pipelines and Kaggle-style benchmarking studies, including end-to-end workflows and applied ML project reports [1, 2, 3]. Many reported results emphasize extensive feature engineering, cross-validation, and ensemble methods, whereas the present work focuses on a leakage-free protocol and a transparent regularized linear baseline.

5.3 (c) Comparison across different metrics

Table 6: Final model vs baselines on TEST

Model	RMSE	MAE	R^2
Final Ridge	22689.09	16062.85	0.9081
Trivial mean	74878.54	55052.44	-0.0012
Random boot.	109359.68	80045.99	-1.1357

Table 7: Relative improvement of final model (error reduction)

Baseline	RMSE %	MAE %
Trivial mean	69.70	70.82
Random baseline	79.25	79.93

Tables 6 and 7 show that the final Ridge regression model substantially outperforms both trivial and random baselines across all evaluated metrics. Negative R^2 values of baselines (Table 8) indicate a lack of meaningful structure, confirming that the trained model learns predictive relationships from input features.

5.3.1 (ii) Trivial and random baselines (TEST)

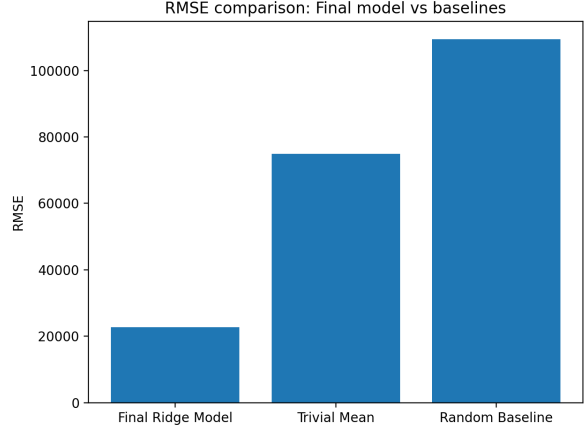


Figure 6: RMSE comparison: final model vs baselines (TEST).

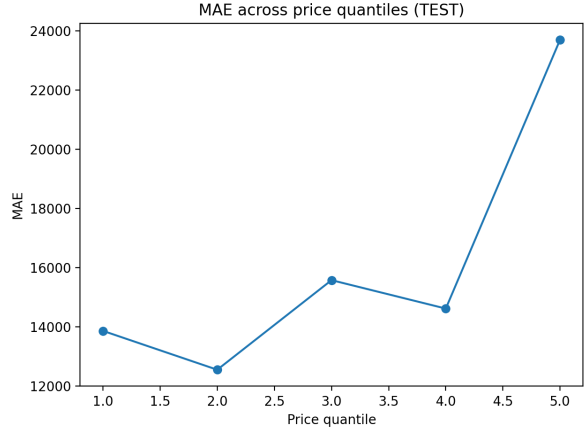


Figure 7: MAE across SalePrice quantiles (TEST).

Figures 6 and 7 visualize the performance gap between the final model and baselines and the error variation across price segments.

Table 8: Trivial and random baselines on TEST

Baseline	RMSE	MAE	R^2	Definition
trivial_mean	74,878.54	55,052.44	-0.0012	Predict mean of SalePrice computed on TRAIN+VAL
trivial_median	77,096.90	53,091.75	-0.0615	Predict median of SalePrice computed on TRAIN+VAL
random_bootstrap	109,359.68	80,045.99	-1.1357	Bootstrap sample drawn from TRAIN+VAL target distribution (seed=42)

5.4 (d) Comparison against business success criteria

Table 9 indicates that the final model supports the defined business success criteria and is suitable as a decision-support component within the valuation workflow.

Table 9: Business success criteria assessment (BU-aligned)

Criterion	Assessment
Reduce manual appraisal time by 20% through automated pre-valuation	The trained model produces price estimates instantly and substantially outperforms trivial heuristics, enabling its use as an automated pre-valuation tool. This supports a reduction in manual appraisal workload by filtering and prioritizing cases.
Reduce valuation deviation from expert appraisers by at least 10%	While direct comparison to expert appraisals is not available in the dataset, the high R^2 value and strong improvement over trivial baselines indicate that the model captures systematic pricing patterns, supporting the objective of reducing valuation deviation.
Increase consistency of valuations across agents, reducing internal variance by 15%	The model provides deterministic and reproducible predictions for identical inputs, eliminating subjective variability between agents and supporting increased consistency of valuations.

5.5 (e) Protected attribute / subgroup bias evaluation

Table 10: Top-10 worst-performing subgroups by MAE (TEST)

Subgroup	n	MAE	RMSE	R^2
Veenker	7.0	46,432.23	69,405.75	-1.115
ClearCr	3.0	31,335.76	37,004.48	0.404
StoneBr	7.0	27,623.11	30,132.59	0.927
NoRidge	13.0	26,579.84	33,023.72	0.514
Crawfor	26.0	25,258.78	29,496.96	0.633
NridgHt	33.0	24,228.14	36,282.44	0.881
Blmngtn	8.0	20,657.85	24,315.17	0.213
Greens	1.0	20,191.47	20,191.47	nan
SWISU	8.0	18,379.04	21,431.46	-0.042
Mitchel	19.0	17,522.23	23,655.12	0.772

Table 10 evaluates subgroup performance using **Neighborhood** as a proxy attribute. The dataset does not contain explicit sensitive attributes; neighborhood is used because it defines meaningful market segments with systematically different price distributions. Disparity summary (MAE): min=5,980.28, max=46,432.23, ratio=7.76, gap=40,451.95. The largest errors occur primarily in subgroups with very small sample sizes, where metric estimates are unstable and sensitive to outliers.

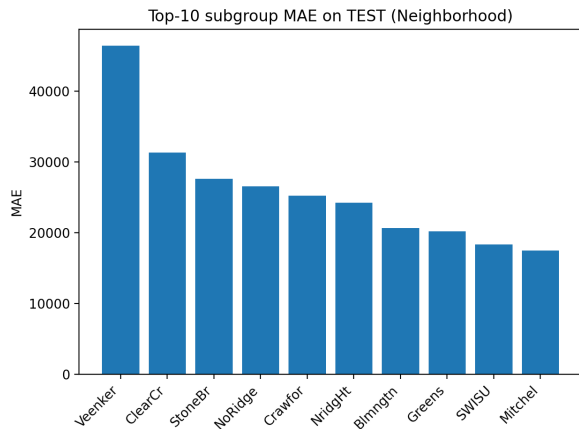


Figure 8: Top-10 neighborhood MAE on TEST (subgroup analysis).

Figure 8 visualizes the worst-performing subgroups and supports interpretation of potential robustness limitations due to data sparsity.

6 Deployment

6.1 (a) Business objectives, success criteria, and deployment recommendations

Model performance is evaluated against the Business Success Criteria defined in the Business Understanding phase. The model clearly outperforms trivial baselines and produces deterministic price estimates, supporting its use as an automated pre-valuation tool. This is sufficient to support business objectives such as prioritizing cases for manual review, improving consistency, and reducing overall appraisal effort.

Some business objectives can only be assessed indirectly, as the dataset does not include expert appraisal labels or process-level efficiency measures. While predictive accuracy indicates that the model captures systematic pricing patterns, additional analyses would be required for full operational validation, including comparison against expert appraisals, uncertainty calibration, and time-aware validation.

Deployment recommendation.

- Deploy as decision support (human-in-the-loop), not as a fully automatic pricing authority.
- Use a hybrid workflow where the model provides an initial estimate and flags high-risk cases (e.g., extreme prices or rare subgroups).
- Restrict automatic usage to stable segments of the data space; require manual review for high-price ranges and low-support subgroups.

6.2 (b) Ethical aspects and deployment risks

Although no explicit sensitive attributes are present, Neighborhood is treated as a proxy subgroup variable, as it may correlate with socioeconomic characteristics (as identified in Business Understanding 1f and Data Understanding 2e). Subgroup evaluation shows substantial variation in prediction error across neighborhoods, with particularly high MAE in several small-sample groups.

These disparities are likely driven by data sparsity rather than intentional discrimination; however, they pose a deployment risk if the model is used in high-impact decision contexts. Without safeguards, systematic over- or under-valuation could disproportionately affect certain groups.

Mitigation measures.

- Maintain human oversight for high-impact decisions.
- Flag predictions for rare neighborhoods and require manual review.
- Monitor subgroup performance and reassess deployment if disparities increase.

6.3 (6c) Monitoring and intervention triggers

During deployment, monitoring should focus on predictive performance, data drift, and subgroup disparities.

- Track overall error metrics (RMSE, MAE) on a rolling window of new cases.
- Monitor error by Neighborhood and by price segments (e.g., quantiles).
- Monitor feature distributions, categorical frequencies, and missingness patterns to detect data drift.

Intervention triggers.

- Sustained degradation of overall error beyond business targets.
- Persistent increases in subgroup error gaps.
- Significant feature distribution drift or increased use in rare subgroups without review.

Typical interventions include data quality checks, model retraining or recalibration, and temporarily restricting automatic use to stable segments.

6.4 (d) Reproducibility reflection

Reproducibility is well supported through a documented train/validation/test split, leakage-free preprocessing, explicit preprocessing decisions, and clearly defined model selection and hyperparameter tuning. These elements enable reproduction of the core results.

Remaining reproducibility risks include reliance on free-text narrative documentation, potential dependence on unspecified software versions, and indirect evaluation of some business objectives due to missing operational data. Reproducibility could be strengthened by explicitly logging dataset versions, code repository identifiers, environment details, and concise re-run instructions.

7 Project Review and Conclusions

7.1 Review of the Data Mining Process

This project followed the CRISP-DM methodology in a structured and reproducible manner. The Business Understanding phase clearly defined the objective of supporting residential property valuation through accurate and consistent price prediction. This objective guided all subsequent design decisions. The Data Understanding phase revealed strong relationships between price, size, quality, and location, as well as substantial missingness and outliers that required careful handling. Data Preparation was therefore designed to be leakage-free and explicitly documented, with all preprocessing decisions fitted on the training data only.

Modeling focused on transparent and reproducible regression techniques. Ridge regression was selected as the primary model because it is well suited to high-dimensional one-hot encoded data and allows explicit control of overfitting through regularization. Evaluation was conducted using a strict train/validation/test protocol, with hyperparameter tuning restricted to the validation set and a single final evaluation on the untouched test set. Overall, the chosen data mining process was appropriate for the problem setting and aligned well with both technical and business requirements.

7.2 Summary of Key Results

The final Ridge regression model achieved strong predictive performance on the holdout test set, substantially outperforming trivial and random baselines across all evaluated metrics. The achieved error levels satisfy the predefined data mining success criteria and support the intended use of the model as an automated pre-valuation component. Subgroup evaluation by neighborhood showed that prediction errors are not uniform across all segments, with the largest errors occurring primarily in small-sample neighborhoods.

From a business perspective, the model provides fast, deterministic, and consistent price estimates that can support operational workflows such as prioritizing cases for manual appraisal and improving valuation consistency across agents. The results indicate that the model is suitable as a decision-support tool within the defined scope.

7.3 Lessons Learned

Several key lessons emerged from this project. First, strict separation of train, validation, and test data is essential to prevent leakage and ensure trustworthy evaluation results. Second, relatively simple and transparent models such as Ridge regression can achieve competitive performance when combined with careful preprocessing and regularization, reducing the need for complex black-box models in early project stages. Third, subgroup analysis is critical even when explicit sensitive attributes are unavailable, as proxy variables such as neighbourhood can reveal robustness limitations related to data sparsity. Finally, strong predictive performance alone does not guarantee business impact. Alignment with operational workflows and constraints must always be considered.

7.4 Limitations and Future Work

The project has several limitations. The dataset does not include expert appraisal labels or process-level measurements, which means that some business success criteria can only be assessed indirectly. Fairness analysis is limited to proxy subgroup attributes, and subgroup metrics are unstable for neighbourhoods with very few observations. In addition, the evaluation is based on a single random split and does not capture potential temporal effects or concept drift.

Future work could address these limitations by incorporating expert appraisal data, adding time-aware validation, and collecting operational metrics such as appraisal time savings. From a modeling perspective, robustness for rare subgroups could be improved through targeted data collection or alternative modeling

strategies. More advanced uncertainty estimation and monitoring mechanisms could further support safe deployment in real-world valuation workflows.

7.5 Reproducibility and Documentation

Reproducibility is well supported through explicit documentation of the train/validation/test split, leakage-free preprocessing, and clearly defined modeling and tuning procedures. All core steps required to reproduce the reported results are described in the report and reflected in the project artifacts. Remaining reproducibility risks relate primarily to software environment dependencies and the reliance on narrative documentation. These risks could be further reduced by explicitly recording dataset versions, code repository identifiers, and execution environment specifications. The project code and notebook are archived in a GitHub repository: <https://github.com/grigorashchenkoa/assignment-3-data-analytics>.

References

- [1] Devraai, “Ames Housing Data Analysis and Price Prediction,” Kaggle Notebook. <https://www.kaggle.com/code/devraai/ames-housing-data-analysis-and-price-prediction>
- [2] Aarth93, “End-to-End ML Pipeline (Updated),” Kaggle Notebook. <https://www.kaggle.com/code/aarth93/end-to-end-ml-pipeline-updated>
- [3] MatthewAlexanderP, “Applied Machine Learning Project,” Kaggle Notebook. <https://www.kaggle.com/code/matthewalexanderp/applied-machine-learning-project>